

The Mathematics of the Thesis
Identifiability, Smile Transport and Finite-Sample Model-Risk Certification
for the JIBAR-to-ZARONIA Derivatives Book

Revised after council review, and re-cut to a three-year programme

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Abstract

This document states every mathematical result the thesis promises, at proposal precision: hypotheses, statement, proof route, failure mode. It is the third version. The first version was put before three reviewers acting as a mathematical statistician, a mathematical finance theorist and a doctoral examiner; their corrections are incorporated and marked in the margin. Two corrections were structural. First, the direction of the identifiability argument in the finance half was reversed: the unidentified correlations contaminate the observed JIBAR smile, not the target compounded-rate smile. Second, the statistics half as first drafted assembled known results and would not carry a mathematical-statistics doctorate; a new theorem on the calibration-conditional coverage error over a mixing class, with the optimal block length of the blocked route as its first part (Theorem 9), is now the spine. The third version corrects Theorem 9 after a literature scan: the blocked rate $N^{-r/(2r+3)}$ of the second version is not minimax in any coverage formulation (Barber and Pananjady 2026; part (ii) below), and the theorem is restated for the calibration-conditional coverage error at confidence δ , where mixing is provably the binding constraint. The final section re-cuts the programme to thirty-six months.

Summary of council corrections

Result	Change forced by review
Lemma 1	Same- g scaling is valid without Dambis–Dubins–Schwarz; linear g and the factor $\Delta/3$ are assumptions, not theorems. Non-decaying vol-of-vol added as Proposition 2; its claimed 17% curvature understatement was wrong (total variance is not the smile) and is corrected to a 0.002 bp effect.
Theorem 3	JIBAR caplet is priced under $\mathbb{Q}^{T+\Delta}$, not $\mathbb{Q}^T$; the in-advance/in-arrears convexity term does not exist and is deleted. Basis correlations enter the JIBAR smile, not the compounded caplet. Two unidentified correlations, not one. Expansion parameter redefined.
Theorem 4	Becomes a corollary of Theorem 3; difference is at the money, not wing-only; rank condition is trivial. Demoted from difficulty 4 to 2–3.
Proposition 5	“Static arbitrage” replaced by “model inconsistency”; exact only for normal SABR with equal volatilities; otherwise leading order in variance. Demoted to proposition, appendix in the three-year plan.
Theorem 6	Coupling cost $(n+1)\beta(b)$, not $2n\beta(b)$; deployment block must be gap-separated; conditional form uses the Beta law of conformal coverage, not DKW; drift penalty is Kolmogorov distance $d_K \leq d_{TV}$, applied after coupling.
Proposition 7	Attribution: the $\sqrt{2LW_1}$ inequality is in print (Ross 2011; Xu et al. 2025, Prop. 1) and is specialised, not claimed. The first version was false (counterexample: point mass versus narrow uniform). Correct bound is $\sqrt{2LW_1}$; plug-in rate $n^{-1/4}$; the dependent-data version of the empirical W_1 bound is currently missing.
Theorem 8	$\delta_{+\infty}$ is not a measure on $\mathbb{R}$; construction replaced; the sharp constant is 1, not $1 - \alpha$; the argument is adversarial, not Le Cam.
First-draft bootstrap theorem	Dropped as a theorem. Empirical comparison with block bootstrap retained.
Theorem 9	New in the second version; corrected in the third. The claimed minimax rate $N^{-r/(2r+3)}$ was false: Barber and Pananjady (2026) obtain the marginal deficit $N^{-r/(r+1)}$ over $\mathcal{B}_r$ without blocking, and the unblocked sample quantile has conditional coverage error $N^{-1/2}$ for every $r > 1$. The deficit $\sqrt{b/N} + Nb^{-(r+1)}$ comes from Theorem 6(ii), not (i). Exact minimiser $(2(r+1))^{2/(2r+3)} N^{3/(2r+3)}$ added. Restated as the minimax calibration-conditional coverage error at confidence δ over $\mathcal{B}_r$, with a mixing-driven lower bound; positioned against Halkiewicz (2026), whose lower bound is drift-driven. This is the doctoral theorem.

1 Setting and notation

Fix a filtered probability space carrying a Brownian motion (W, Z) with $d\langle W, Z \rangle_t = \rho dt$, and a third Brownian motion W^B with $d\langle W^B, W \rangle_t = \rho_{BF} dt$, $d\langle W^B, Z \rangle_t = \rho_{B\alpha} dt$. Let $P(t, U)$ denote the zero-coupon bond price and $\mathbb{Q}^U$ the U -forward measure.

Compounded rate. For an accrual window $[T, T + \Delta]$ with daily fixings r_i and day-count fractions δ_i ,

$$R(T, T + \Delta) = \frac{1}{\Delta} \left(\prod_i (1 + r_i \delta_i) - 1 \right) \approx \frac{1}{\Delta} \left(\exp \left(\int_T^{T+\Delta} r_u du \right) - 1 \right). \quad (1)$$

The continuous approximation is used throughout; it makes $1 + \Delta F_t = P(t, T)/P(t, T + \Delta)$ exact and differs from the daily product at second order in $r_i \delta_i$, below a basis point. [Council correction:

state once, keep continuous form]

Forwards. The forward compounded rate $F_t := \mathbb{E}^{\mathbb{Q}^{T+\Delta}}[R(T, T+\Delta) \mid \mathcal{F}_t]$ is a $\mathbb{Q}^{T+\Delta}$ -martingale. The JIBAR forward L_t for the same window is fixed at T and *paid at* $T + \Delta$; it is therefore also a $\mathbb{Q}^{T+\Delta}$ -martingale, and so is the basis $B_t := L_t - F_t - s$, where s is the fixed credit adjustment spread. In particular $\mathbb{E}^{\mathbb{Q}^{T+\Delta}}[B_T] = B_0 = L_0 - F_0 - s$ is read off the two curves. [Council correction: both caplets live under $\mathbb{Q}^{T+\Delta}$; no in-advance/in-arrears convexity term]

Decay and clock. Let g be deterministic and bounded with $g(t) = 1$ for $t \leq T$ and $g(t) \downarrow 0$ on $(T, T + \Delta]$, and set $\tau(t) := \int_0^t g(u)^2 du$.

Assumption 1 (Linear decay). $g(t) = (T + \Delta - t)/\Delta$ on $(T, T + \Delta]$, so that $\tau(T + \Delta) = T + \Delta/3$.

[Council correction: linear g is exact only when the instantaneous forward-rate volatility is flat in maturity; generically $\tau(T + \Delta) = T + c_g \Delta$ with $c_g \in (0, 1)$. Every result below inherits this assumption.]

Dynamics. Under $\mathbb{Q}^{T+\Delta}$,

$$dF_t = g(t) \alpha_t F_t^\beta dW_t, \quad d\alpha_t = g(t) \nu \alpha_t dZ_t, \quad dB_t = \eta dW_t^B. \quad (2)$$

The dimensionless basis scale is $\varepsilon := \eta/(\alpha_0 F_0^\beta)$. [Council correction: $\eta\sqrt{T}$ is a rate, not a small parameter]

Scores. For a fixed valuation-and-hedging policy π , the hedging error over a block of b consecutive rebalancing periods is the *score* S . Calibration scores have law P , deployment scores law Q . F_P, F_Q are their distribution functions, $d_K(P, Q) = \sup_x |F_P(x) - F_Q(x)| \leq d_{TV}(P, Q)$ the Kolmogorov distance.

2 Finance results

Lemma 1 (Time-changed SABR; Lyashenko–Mercurio 2019, Remark in Willems 2020). *Under (2) with the same g on both diffusion coefficients, $(F_t, \alpha_t) = (\tilde{F}_{\tau(t)}, \tilde{\alpha}_{\tau(t)})$ where $(\tilde{F}, \tilde{\alpha})$ is constant-parameter SABR($\alpha_0, \beta, \rho, \nu$). Hence*

$$P(0, T + \Delta) \mathbb{E}^{\mathbb{Q}^{T+\Delta}}[(F_{T+\Delta} - K')^+] = P(0, T + \Delta) C^{\text{SABR}}(F_0, K', \tau(T + \Delta); \alpha_0, \beta, \rho, \nu), \quad (3)$$

with the Hagan approximation carrying its usual $O(\alpha^2 \tau)$ error.

Proof. Because τ is deterministic, $(\int_0^t g dW, \int_0^t g dZ)$ is a correlated Brownian pair in the clock τ ; substitute. [Council correction: no Dambis–Dubins–Schwarz needed; the same- g condition is what makes the substitution exact] $\square$

Proposition 2 (Non-decaying vol-of-vol; Willems 2020, Theorems 4.1–4.2; target). *If instead $d\alpha_t = \nu \alpha_t dZ_t$ (vol-of-vol does not run off, the physically defensible choice), then in the clock τ the vol-of-vol is $\nu/g(\tau^{-1}(u))$ and no closed form exists.*

(i) (Total variance.) *The total vol-of-vol variance is $\nu^2(T + \Delta)$ rather than $\nu^2(T + \Delta/3)$: an excess of $\frac{2}{3}\nu^2\Delta$, which is 15.4% of $\nu^2\tau^*$ and 16.7% of ν^2T for $T = 1y$, $\Delta = 3m$. This quantity is the variance of $\log \alpha_{T+\Delta}$; it does not by itself determine the smile.*

(ii) (Smile-relevant effective parameters.) At leading order the Hagan–Lesniewski–Woodward effective parameters, computed in calendar time with $du = g^2 dt$ and $\int_T^{T+\Delta} g^{2k} dt = \Delta/(2k+1)$, satisfy

$$\frac{\nu_{\text{eff}}^2}{\nu^2} = 1 + \frac{2\Delta^3/567}{\tau^{*3}/3} = 1.00013, \quad \frac{\rho_{\text{eff}}\nu_{\text{eff}}}{\rho\nu} = 1 + \frac{\Delta^2/90}{\tau^{*2}/2} = 1.0012$$

for $T = 1y$, $\Delta = 3m$, so the two variants agree to about 0.002 bp at ± 50 bp from the money. Lemma 1 does not understate smile curvature in any material way.

[Council correction: the second version claimed a 17% curvature understatement by identifying curvature with total vol-of-vol variance; the forward-measures referee showed the smile-relevant effective parameters differ by 0.013% and 0.12%, confirmed numerically by the rates code. Whether the leading-order weights coincide with Hagan et al. (2002), Appendix B, is a librarian task.]

Remark. Neither result is a contribution. Lemma 1 is cited; Proposition 2 is a two-page derivation in Chapter 2 and fixes which variant the empirical chapters use.

Theorem 3 (Identified and unidentified components of the smile map; [target](#)). *Let (2) hold with Assumption 1, fix β , and suppose $\nu \neq 0$. Denote by $\sigma_L^N(K, T)$ the observed JIBAR normal implied volatility and by $\sigma_F^N(K', T + \Delta)$ the compounded-rate normal implied volatility at the fallback strike $K' = K - s$. Then*

$$\sigma_F^N(K', T + \Delta) = \Phi_0(K'; \sigma_L^N(\cdot, T), s, B_0, \Delta) - \rho_{BF} \eta \Phi_1(K') - \rho_{B\alpha} \eta \nu \Phi_2(K') + O(\varepsilon^2), \quad (4)$$

where

1. Φ_0 is the SABR smile at clock $T + \Delta/3$ whose triple (α_0, ρ, ν) is fitted to $\sigma_L^N(K' + s + B_0, T)$ at clock T : a strike shift by $s + B_0$ and the time change of Lemma 1;
2. $\Phi_1(K') = 1 + O(K' - F_0)$: a parallel level shift;
3. $\Phi_2(K') \propto (K' - F_0)$: a skew tilt.

Consequently: the curvature parameter ν and the map Φ_0 are identified from the observable set $\mathcal{S} = \{\sigma_L^N(\cdot, T)\} \cup \{B_0\} \cup \{s\}$; the pair $(\eta\rho_{BF}, \eta\rho_{B\alpha})$ is not identified at $O(\varepsilon)$, and $(\eta, \rho_{BF}, \rho_{B\alpha})$ is not identified at $O(\varepsilon^2)$. In particular the successor at-the-money volatility is identified only within an interval of half-width η .

[Council correction: the first draft had the unidentified correlation entering the compounded caplet; it is the other way round. The compounded caplet depends only on $(\alpha_0, \beta, \rho, \nu)$. The JIBAR smile is a contaminated view of that triple, because $L = F + B + s$ and the basis noise shifts its level (through ρ_{BF}) and tilts its skew (through $\rho_{B\alpha}\nu$). The convexity term was deleted because both instruments pay at $T + \Delta$.]

Proof route. Under $\mathbb{Q}^{T+\Delta}$ write $L = F + B + s$ and expand the JIBAR caplet in ε : the level of σ_L^N is σ_F^N -at-clock- T plus $\rho_{BF}\eta$ at first order (variance of a sum), the skew acquires $\rho_{B\alpha}\eta\nu$ from $\text{Cov}(B, \alpha)$, and the curvature is unaffected at $O(\varepsilon)$. Invert: fitting SABR to σ_L^N returns $(\alpha_0 - \rho_{BF}\eta + \dots, \rho + c\rho_{B\alpha}\eta\nu/\alpha_0 + \dots, \nu)$. Push the fitted triple through the time change to obtain Φ_0 ; the residual terms are Φ_1, Φ_2 .

Failure mode. If the basis has mean reversion comparable to $1/T$, the $O(\varepsilon)$ coefficients acquire T -dependence and the interval half-width becomes $\eta h(\kappa T)$ for an explicit h . The statement survives with a modified constant.

Theorem 4 (Two models; corollary of Theorem 3). *For any (v_1, v_2) with $|v_i| \leq \eta$ there exist models $\mathcal{M}_\pm$ with $(\eta\rho_{BF}, \eta\rho_{B\alpha}) = \pm(v_1, v_2)$ that match every JIBAR caplet to $O(\varepsilon^2)$ and match B_0 exactly, while their compounded-rate at-the-money volatilities differ by $2v_1$ and their skews by $2v_2\nu$, at every strike.*

Proof. Flip the sign of $(\rho_{BF}, \rho_{B\alpha})$ and re-solve (α_0, ρ) so that the first-order level and skew of σ_L^N are unchanged. The Jacobian $\partial(\text{level, skew})/\partial(\alpha_0, \rho)$ is nonsingular for $\nu \neq 0$, so the re-solve exists. Read the differences off (4). [Council correction: rank is immediate, not a research question; difference is at the money, not wing-only] $\square$

Remark (Parameter count). With m expiries and β fixed: observables are $3m$ smile features (level, skew, curvature per expiry; additional strikes test SABR shape and carry η only at $O(\varepsilon^2)$) plus m basis points, which pin $B_{0,i}$ and hence strike shifts but no smile parameters. Unknowns are $3m$ SABR parameters plus $(\eta, \rho_{BF}, \rho_{B\alpha})$ if shared across expiries. Deficit: 3 (two at $O(\varepsilon)$). If the basis parameters are per-expiry, the deficit is $3m$. Historical basis data estimate the nuisance under the physical measure, not the pricing measure; that estimate is the ambiguity set that Theorem 6 takes as input.

Proposition 5 (Tenor consistency; *target*, appendix in the three-year plan). *For $t \leq T_1 < T_2 < T_3$, $(1 + \Delta_{13}F_t^{13}) = (1 + \Delta_{12}F_t^{12})(1 + \Delta_{23}F_t^{23})$ exactly (ratios of zero-coupon bonds).*

- (a) *If $\beta = 0$, g linear and (W, Z) common, the induced R_{13} is time-changed normal SABR with linear decay if and only if $\alpha_{12} = \alpha_{23}$; then $\alpha_{13} = \alpha_{12}$, (ρ, ν) are inherited, and*

$$\Delta_{13}^2 \tau_{13} = \Delta_{12}^2 \tau_{12} + \Delta_{23}^2 \tau_{23} + 2\Delta_{12}\Delta_{23}c, \quad c = T_1 + \Delta_{12}/2,$$

exactly; the clocks compose.

- (b) *Otherwise the displayed relation is the variance-matched projection, valid to leading order in $\alpha^2\tau$, and the constraints tying (ρ, ν) across tenors are approximate.*
- (c) *Violation of (a)–(b) means no model in the class calibrates jointly to the 3M and 6M smiles. It is not a static arbitrage: $(R_{13} - K)^+$ is not spanned by 3M caplets. [Council correction: a model-free statement would need Hobson–Laurence–Wang-type bounds for an option on a product with given marginals, which is a different theorem and is not claimed]*

3 Statistics results

Assumption 2 (Mixing and separation). The market state process that generates both the fitting sample and the per-period hedging errors of the fixed policy π is β -mixing (the error process of a policy fitted on a random sample does not inherit the coefficients of the state process by itself; the assumption is placed on the state process) with coefficients $\beta(k)$ (non-stationary definition: supremum over time origins) and stationary on the calibration window. The fitting sample $\mathcal{D}_{\text{fit}}$, the n calibration blocks of length b , and the deployment block are pairwise separated by gaps of length at least b . [Council correction: the deployment gap is required; without it S_0 is not controlled at all]

Let $k := \lceil (1 - \alpha)(n + 1) \rceil$, $\hat{q} := S_{(k)}$ the k -th order statistic of the calibration scores, and $b_{n,k}(\delta) := \text{Beta}^{-1}(\delta; k, n + 1 - k)$ the δ -quantile of the $\text{Beta}(k, n + 1 - k)$ law.

Theorem 6 (Coverage under mixing and drift; *target*). *Under Assumption 2, with calibration scores $S_1, \dots, S_n \sim P$ marginally and deployment score $S_0 \sim Q$:*

- (i) (Marginal.) $\mathbb{P}(S_0 \leq \hat{q}) \geq 1 - \alpha - d_K(P, Q) - (n + 1)\beta(b)$.
- (ii) (Conditional.) *For any $\delta, \delta' \in (0, 1)$, with probability at least $1 - \delta - n\beta(b)$ over $(\mathcal{D}_{\text{fit}}, S_{1:n})$,*

$$\mathbb{P}(S_0 \leq \hat{q} \mid \mathcal{D}_{\text{fit}}, S_{1:n}) \geq b_{n,k}(\delta) - d_K(P, Q) - \beta(b)/\delta',$$

the last term holding on an event of probability at least $1 - \delta'$. Moreover $b_{n,k}(\delta) \geq 1 - \alpha - \sqrt{\log(1/\delta)/(2n)}$. (The normal approximation $b_{n,k}(\delta) \approx 1 - \alpha - \sqrt{2\alpha(1 - \alpha) \log(1/\delta)/n}$ is a large- n heuristic that bounds in neither direction; at $n = 25$, $\alpha = 0.1$, $\delta = 0.05$ the exact quantile is 0.8239 against 0.7531 for the Gaussian tail and 0.8387 for the moment-matched normal. It is not part of the statement.)

No assumption is made that the model underlying π is correct.

[Council correction: coupling cost is $(n+1)\beta(b)$ (Yu 1994, Lemma 4.1: $(m-1)\beta(b)$ for m blocks, plus independence from $\mathcal{D}_{\text{fit}}$ and the deployment block); the conditional statement cannot be obtained from a joint total-variation bound and uses the β -mixing definition with Markov's inequality; the conformal coverage is $\text{Beta}(k, n+1-k)$ -distributed, the Beta form of Vovk (2012, Proposition 2b, stated there in binomial form), so DKW is a corollary, not the tool; the drift penalty is Kolmogorov distance because only half-lines are evaluated, applied to the fixed set $(-\infty, \hat{q}^*]$ after coupling has made S_0^* independent of $\hat{q}^*$.]

Proof route. (i) Sequential Berbee coupling replaces the $n+1$ separated blocks and $\mathcal{D}_{\text{fit}}$ by independent copies $S_{0:n}^*$ on an event of probability $\geq 1 - (n+1)\beta(b)$. (ii) On that event, conditional on π , the calibration copies are i.i.d. P and $F_P(\hat{q}^*) \sim \text{Beta}(k, n+1-k)$, so $\mathbb{P}(S_0^* \leq \hat{q}^*) \geq 1 - \alpha$ if $S_0^* \sim P$. (iii) For $S_0^* \sim Q$ and the fixed half-line $(-\infty, \hat{q}^*]$, $|Q(S \leq \hat{q}^*) - P(S \leq \hat{q}^*)| \leq d_K(P, Q)$. Sum.

Remark. $\beta(b)$ is not estimable from data (Adams–Nobel-type impossibility). The certificate is conditional on an *assumed* mixing class $\{\beta(k) \leq Ck^{-r}\}$, and the thesis says so. [Council correction: added]

Proposition 7 (Drift penalty bounded by W_1 ; known: Ross 2011, Xu et al. 2025 Prop. 1). *If Q has a density bounded by L (no assumption on P), then for all x , $F_P(x) - F_Q(x) \leq \sqrt{2LW_1(P, Q)}$. If both have densities bounded by L , $d_K(P, Q) \leq \sqrt{LW_1(P, Q)}$. Also $F_P(x) - F_Q(x) \leq LW_\infty(P, Q)$.*

Proof. Let $D = F_P - F_Q$. If $D(x_0) = d > 0$ then, since F_P is non-decreasing and F_Q is L -Lipschitz, $D(x) \geq d - L(x - x_0)$ on $[x_0, x_0 + d/L]$, whence $W_1(P, Q) = \int |D| \geq d^2/(2L)$. The W_∞ bound uses the monotone coupling: $F_P(x) \leq F_Q(x + W_\infty) \leq F_Q(x) + LW_\infty$. $\square$

[Council correction: the first version claimed $|F_P - F_Q| \leq LW_1$, which is false: $P = \delta_0$, $Q = U[0, 1/L]$ has $W_1 = 1/(2L)$ and CDF gap 1. With the square-root form the plug-in penalty decays at $n^{-1/4}$, not $n^{-1/2}$. The i.i.d. rate $\mathbb{E}W_1(P_n, P) \leq n^{-1/2}J_1(P)$ is Bobkov–Ledoux (2019) in $d = 1$; deployment scores are dependent, so a β -mixing version (Dedecker–Merlevède type) is needed and is part of Theorem 9(v). L is an assumption on the unobserved Q and is stated as such.]

Theorem 8 (Sharpness of the drift penalty). *Let T be any threshold procedure independent of the deployment score, and $C_T(R) := \mathbb{P}_{S \sim R}(S \leq T)$. For every P with continuous distribution function and every $\varepsilon \in (0, 1)$:*

- (a) (Universal.) $\inf_{Q: d_{\text{TV}}(P, Q) \leq \varepsilon} C_T(Q) \geq C_T(P) - \varepsilon$.
- (b) (Attained.) Let Q_M be P with its bottom- ε mass on $(-\infty, q_\varepsilon(P)]$ moved to the point M . Then $d_{\text{TV}}(P, Q_M) = d_K(P, Q_M) = \varepsilon$ and

$$C_T(Q_M) \leq C_T(P) - \varepsilon + \varepsilon \mathbb{P}(F_P(T) < \varepsilon) + \varepsilon \mathbb{P}(T \geq M).$$

For split conformal, $\mathbb{P}(F_P(T) < \varepsilon) = \mathbb{P}(\text{Beta}(k, n+1-k) < \varepsilon)$, exponentially small for $\varepsilon < 1 - \alpha$. Hence the constant 1 on d_K in Theorem 6 is sharp.

Proof. (a) is the definition of total variation applied to the event $\{S \leq T\}$ conditional on T . (b) Under Q_M , mass ε that P placed below every plausible threshold now sits at M ; if $T < M$ and $F_P(T) \geq \varepsilon$, coverage falls by exactly ε . The two correction terms cover the complementary events. $\square$

[Council correction: $\delta_{+\infty}$ is not a probability measure on $\mathbb{R}$. The far-mass construction $(1 - \varepsilon)P + \varepsilon\delta_M$ gives only $(1 - \varepsilon)C_T(P)$, which is weaker and sharp only for degenerate procedures. Moving mass from the bottom of P to a far point is what attains the constant 1. Since Q is never observed, this is a direct

adversarial construction, not Le Cam's method; Le Cam enters only in Theorem 9(iv) where deployment scores are partially observed.]

Theorem 9 (Calibration-conditional coverage error over a mixing class: optimal blocking and minimax rate; *target*, the doctoral theorem). *Fix the mixing class $\mathcal{B}_r := \{\beta(k) \leq Ck^{-r}\}$, $r > 1$, and a calibration record of N periods, stationary on the record with continuous marginal score law P ; write F_N for the empirical distribution function of the N scores. A threshold procedure is any measurable $T = T(S_{1:N})$. Its conditional coverage error is $\mathcal{E}(T) := |F_P(T) - (1 - \alpha)|$, the error in the coverage $F_P(T)$ received by a deployment score independent of the record; the deployment gap and its coupling cost are handled as in Theorem 6 and are excluded here, as is drift. The error is two-sided because the one-sided deficit is degenerate over all procedures ($T = +\infty$ has deficit zero); over-coverage is the efficiency loss. The coverage error at confidence δ is $\mathcal{E}_\delta(T) := \inf\{\varepsilon : \mathbb{P}(\mathcal{E}(T) > \varepsilon) \leq \delta\}$ and the minimax value is $\mathcal{E}_\delta^*(N, r) := \inf_T \sup_{\mathcal{B}_r} \mathcal{E}_\delta(T)$.*

(i) (Optimal block length for the blocked route; provable.) Blocks of length b with gaps b give $n = \lfloor N/(2b) \rfloor$ blocks. By Theorem 6(ii), the Beta($k, n + 1 - k$) law of $F_P(\hat{q})$ on the coupling event integrated over the confidence level, the expected conditional shortfall of $\hat{q} = S_{(k)}$ obeys

$$\mathbb{E}[(1 - \alpha - F_P(\hat{q}))^+] \leq \frac{1}{2\sqrt{n+2}} + (n+1)\beta(b) \leq \sqrt{\frac{b}{2N}} + C N b^{-(r+1)} \quad (2 \leq b \leq N/2),$$

so that, up to the constants $1/\sqrt{2}$ and C , the bound is $f(b) := \sqrt{b/N} + N b^{-(r+1)}$. The exact minimiser of f is

$$b^* = (2(r+1))^{2/(2r+3)} N^{3/(2r+3)}, \quad f(b^*) = \frac{2r+3}{2(r+1)} (2(r+1))^{1/(2r+3)} N^{-r/(2r+3)},$$

with $f(b^*) = 1.51 N^{-r/(2r+3)}$ and $b^* = 23.9$ at $r = 2$, $N = 500$; the bare balance $\sqrt{b/N} = N b^{-(r+1)}$ gives $N^{3/(2r+3)} = 14.3$ and is not the minimiser.

(ii) (The blocked rate is not minimax; provable.) Since $|\text{Cov}(\mathbf{1}\{S_0 \leq x\}, \mathbf{1}\{S_k \leq x\})| \leq \alpha(k) \leq \beta(k)$, $\text{Var } F_N(x) \leq (\frac{1}{4} + 2C\zeta(r))/N$ for every x , and the unblocked sample quantile $T_N := S_{(\lfloor (1-\alpha)N \rfloor)}$ of the whole record satisfies, for every $\varepsilon > 0$,

$$\mathbb{P}(|F_P(T_N) - (1 - \alpha)| > \varepsilon) \leq \frac{\frac{1}{2} + 4C\zeta(r)}{N\varepsilon^2}, \quad \text{hence} \quad \mathcal{E}_\delta(T_N) \leq \sqrt{\frac{\frac{1}{2} + 4C\zeta(r)}{N\delta}}.$$

I.i.d. scores lie in every $\mathcal{B}_r$, and the two-point argument for them gives $\mathcal{E}_\delta^(N, r) \geq c_\delta N^{-1/2}$. At fixed δ the minimax rate over $\mathcal{B}_r$ is therefore exactly $N^{-1/2}$, attained without blocking; the blocked rate $N^{-r/(2r+3)}$ of (i) is slower by the factor $\sqrt{b^*}$. Blocking buys the exact Beta law and a certificate whose only dependence on (C, r) is the failure probability $(n+1)\beta(b)$; it does not buy rate.*

(iii) (Mixing-driven lower bound at high confidence; *target*.) There are $c_r, \delta_0 > 0$ and, for every N , a stationary process in $\mathcal{B}_r$ (regime-renewal construction: i.i.d. regime labels with score laws $P_1 \neq P_2$, held for i.i.d. durations with tail $k^{-(r+1)}$, so that $\beta(k) \leq \mathbb{P}(\text{no renewal in } k \text{ steps}) \asymp k^{-r}$) such that for every threshold procedure T and every $\delta \in [N^{-r}, \delta_0]$,

$$\mathbb{P}(\mathcal{E}(T) \geq c_r (N^r \delta)^{-1/(r+1)}) \geq \delta.$$

Together with the Fuk–Nagaev inequality for bounded strongly mixing sequences (Rio 2017, Ch. 6) applied to F_N at the two quantiles $q_{1-\alpha \mp \varepsilon}(P)$, this gives

$$\mathcal{E}_\delta^*(N, r) \asymp \max \left\{ N^{-1/2} g_r(\delta), (N^r \delta)^{-1/(r+1)} \right\}, \quad \sqrt{\log(1/\delta)} \lesssim g_r(\delta) \lesssim \delta^{-1/r},$$

and the mixing term binds exactly when $\delta < N^{-(r-1)/2}$ (at $N = 500$, $r = 2$: $\delta < 0.045$, so every certificate at the usual confidence levels sits in the mixing-bound regime). The same construction gives an expected shortfall of order $N^{-r/(r+1)}$, the rate that Barber and Pananjady (2026) prove for split conformal; part (iii) extends the lower bound to all threshold procedures, which they leave open.

- (iv) (Partially observed drift; *target*.) When m deployment scores are additionally observed, the minimax certifiable deficit over $\{Q : d_K(P, Q) \leq \varepsilon\}$ is obtained adaptively in unknown ε by Le Cam’s two-point method on the observed deployment sample.
- (v) (Dependent empirical W_1 ; cited and specialised.) The dependent analogue of the Bobkov–Ledoux bound, $\mathbb{E} W_1(Q_m, Q) \lesssim m^{-1/2}$ under summable dependence with a moment condition on Q , is Dedecker and Merlevède (2017). The thesis specialises it to $\beta \in \mathcal{B}_r$, $r > 1$, with an explicit c_r , closing the gap in Proposition 7. This part is not claimed as new.

[Council correction: third version. The second version stated (ii) as “no threshold procedure achieves deficit smaller than $cN^{-r/(2r+3)}$; hence the rate is minimax over $\mathcal{B}_r$ ”. That is false in every coverage formulation: marginally, Barber and Pananjady (2026, Cor. 1–2) reach $N^{-r/(r+1)}$ without blocking; conditionally, the unblocked sample quantile reaches $N^{-1/2}$ for all $r > 1$ by the covariance inequality, and over all procedures the one-sided deficit is degenerate. The second version also derived $\sqrt{b/N} + Nb^{-(r+1)}$ “by Theorem 6(i)”, which has no $\sqrt{b/N}$ term; the concentration term exists only in the conditional form (ii). The bare balance was quoted as b^* ; the exact minimiser carries $(2(r+1))^{2/(2r+3)}$ (code agent, 2026-09-09). The old part (iii) is Dedecker–Merlevède (2017) and is downgraded to cited.]

Remark (Decision: why the spine is the calibration-conditional error). Three options were weighed. (a) Keep marginal coverage and target the Barber–Pananjady open question at rate $N^{-r/(r+1)}$: rejected as the spine, because over all procedures the marginal deficit is degenerate without an efficiency constraint, and adding one turns the question into (b); the blocked marginal bound of Theorem 6(i) is dominated by their unblocked bound and is retained only as the route to (ii). (b) Restate for the calibration-conditional coverage error at confidence δ : chosen. The concentration term is real there, the two-sided error is non-degenerate over all procedures, the fixed- δ minimax rate is settled by part (ii), and the high-confidence regime of part (iii) is where the mixing exponent is provably the binding constraint, which is the object a model-risk certificate needs. (c) Shift the spine to dependent W_1 plus partially observed drift: rejected, because Dedecker and Merlevède (2017) already give the dependent W_1 rate and the drift part is an adaptation of Le Cam’s method. Positioning: Halkiewicz (2026) proves an optimal calibration window with a Le Cam lower bound for rolling-origin conformal under local stationarity and α -mixing; his hardness is drift-driven and his class carries no β -mixing exponent, no blocking and no conditional coverage. Theorem 9 is not the first optimal-design result for conformal prediction under dependence and must not be described as such; its claim is the mixing-driven minimax statement (iii) over $\mathcal{B}_r$ for conditional coverage, and the honest accounting (ii) of what blocking does and does not buy.

Proof route. (i) $\mathbb{E}[(1 - \alpha - B)^+] \leq \mathbb{E}|B - k/(n+1)| \leq \text{sd}(B) \leq 1/(2\sqrt{n+2})$ for $B \sim \text{Beta}(k, n+1-k)$, since $k/(n+1) \geq 1 - \alpha$; add the coupling failure $(n+1)\beta(b)$ of Theorem 6; differentiate f . (ii) $F_P(T_N) < 1 - \alpha - \varepsilon$ forces $F_N(q_-) - F_P(q_-) \geq \varepsilon$ at $q_- = q_{1-\alpha-\varepsilon}(P)$, and $F_P(T_N) > 1 - \alpha + \varepsilon$ forces $F_N(q_+) - F_P(q_+) < -\varepsilon$; Chebyshev with the covariance bound at each point; the i.i.d. lower bound is the two-point argument on Bernoulli data with $\text{KL} \asymp N\varepsilon^2$. (iii) A holding interval of length $m \asymp (N/\delta)^{1/(r+1)}$ meets the record with probability $\asymp Nm^{-(r+1)} \asymp \delta$ and shifts F_N at the quantile by $\asymp (m/N)d_K(P_1, P_2)$; to pass from the sample quantile to all procedures, compare two regime-renewal processes whose mixture weights differ by $\asymp m/N$ and whose N -sample laws are close in total variation; Fuk–Nagaev (Rio 2017, Thm 6.2 form: a Gaussian-type term $(1 + N\varepsilon^2/(r\sigma^2))^{-r/2}$ plus a polynomial term $N^{-r}\varepsilon^{-(r+1)}$) for the upper bound. (iv) Two-point reduction on the deployment sample. (v) Specialise Dedecker–Merlevède to blocked scores under $\mathcal{B}_r$ with the Berbee coupling of Theorem 6.

Failure mode. (i) and (ii) are elementary and cannot fail. In (iii) the two-point reduction over all procedures may lose a logarithmic factor or hold only over the regime-renewal subclass of $\mathcal{B}_r$; the δ -dependence g_r between $\sqrt{\log(1/\delta)}$ and $\delta^{-1/r}$ is open and is stated as open. Either outcome is publishable; the fixed- δ rate is settled either way.

Remark (On the first-draft bootstrap theorem). The comparison with the stationary block bootstrap is no longer stated as a theorem. Bootstrap coverage rates for quantiles under mixing require Edgeworth expansions with unknown constants, and “the only method with a provable finite-sample level” is a comparison of assumptions, not a result. The comparison is retained as an empirical section on USD and ZAR data. [Council correction: dropped]

4 Difficulty after correction

Result	Kind	Deepest tool	Difficulty (1–5)	Carries the doctorate
Lemma 1	cited	deterministic time change	1	no
Prop. 2	derivation	HLW effective parameters	2	no
Thm 3	expansion, error bound	forward measure, SABR inversion	3	finance paper
Thm 4	corollary	nonsingular Jacobian	2	no
Prop. 5	algebra	variance matching	2–3	appendix
Thm 6	theorem	Berbee coupling, Beta law, d_K	3	supports
Prop. 7	proposition	Lipschitz CDF argument	2	supports
Thm 8	theorem	adversarial construction	2	supports
Thm 9	theorem	regime-renewal lower bound, Fuk–Nagaev under mixing	4	yes: the spine

Not needed anywhere: Malliavin calculus, rough paths, G-expectation, optimal transport, reflected BSDEs, deep-learning theory.

5 The three-year programme

Principles. The statistics spine is proved first, in the candidate’s home field, while finance prerequisites are learned in parallel: Theorem 6 needs only a fixed policy, and the market convention (strike shift, volatility scaling) supplies one without Theorem 3. The USD LIBOR-to-SOFR transition is the empirical spine, because the target smile became observable there and every input is on the university terminal; the South African book is a coda, because by month 30 it will have at most twenty-four monthly hedging-error observations and its option surfaces need a sponsor who may not sign.

Cuts. Proposition 5 to an appendix, stated with the exact case (a) proved and (b) as a conjecture. The first-draft bootstrap theorem to an empirical section. The standalone factual-record note folded into Chapter 1 at three thousand words. The standalone time-changed-SABR note becomes Chapter 2 preliminaries. The South African chapter shrinks to five thousand words.

Minimal passing thesis. Theorems 6, 8 and 9 proved; the USD certificate test with honest coverage; Theorem 3 with the $O(\varepsilon)$ identified part and Theorem 4 demonstrated numerically on USD; the South African coda on public data if no sponsor. This passes because the spine is proved, sharp, rate-optimal, and tested where the truth was observable.

Months	Activity	Milestone
1–4	Stochastic calculus and forward measures; Barber et al. (2023), Oliveira et al. (2024), Barber and Pananjady (2026), Vovk (2012), Rio (2017, Ch. 6) line by line	Reading report
3–6	Theorem 8 proved; Theorem 6 route drafted with the Beta form	Thm 8 done (m5)
5–9	USD multi-curve bootstrap and time-changed SABR caplet code; Lemma 1, Prop. 2	Working code
7–10	Theorem 6 full proof; Proposition 7	Thm 6 done (m10)
9–11	Proposal defence: Chapters 1–2 drafts, Theorems 6, 8 proved, Theorem 9 stated	Faculty approval (m11)
10–12	USD data assembled; ZAR sponsor decision	Data agreement or public-data fallback (m12)
10–18	Theorem 9: (i)–(ii) proved (m12), regime-renewal lower bound (iii) (m16), Fuk–Nagaev upper bound, (iv)–(v) (m18)	Thm 9 done (m18)
12–17	USD certificate test with optimal blocking; bootstrap and traffic-light comparison	Chapter 6 part one
14–20	Theorem 3 expansion; Theorem 4 construction, with finance co-supervisor	Chapter 4 draft (m20)
18–22	Paper 1: Theorems 6, 8, 9, Prop. 7, USD test	Submitted, <i>EJS</i> or <i>Bernoulli</i> (m22)
20–24	USD identifiability validation: predicted SOFR smile and $\pm\eta$ interval against the observed smile	Chapter 6 complete (m24)
24–28	Paper 2: Theorems 3, 4, USD identifiability	Submitted, <i>Quantitative Finance</i> (m28)
26–30	ZAR coda: JIBAR surface to Dec 2026, converted-book P&L to m30, reserve decomposition	Chapter 7 draft (m30)
30–34	Full write-up; Chapters 1, 8, appendices; internal review	Complete draft (m34)
34–36	Corrections, submission	Thesis submitted (m36)

Chapters and words. 1 Introduction and the transition (5k); 2 Preliminaries (8k); 3 Certification (10k); 4 Sharpness and optimal blocking (6k); 5 Identifiability (10k); 6 USD validation (10k); 7 South African coda (5k); 8 Conclusions (3k); appendices. About 57 thousand words.

Papers. Two, not three. Paper 1 is the doctoral paper. The South African material becomes a practitioner note to the Market Practitioners Group.

Probability of submission. Examiner’s estimate: about 60% by month 36; about 85% by month 42 with the customary extension. The plan passes without ZAR data and without Theorem 3 being clean. It does not pass without Theorem 6 by month 10–12, Theorem 9 by month 18, and the USD test by month 17.

Ranking under three years. Three years widens this topic’s lead over the alternatives the council reviewed. Conformal deep hedging on index options needs a year of training infrastructure before any theorem. Certified superhedging under volatility uncertainty needs G-expectation prerequisites and carries difficulty-5 theorems that can fail outright. This topic’s spine uses tools half-known already, and its finance half, which is the schedule risk, is exactly what the re-cut trims.

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